Green Home Score and Retrofit Recommendation Engine

1 Challenge

Many customers want an energy-efficient home but do not know where to start. Build an AI system that analyses property characteristics, energy consumption and existing home features to:

1. calculate a Green Home Score;
2. identify homes with poor energy efficiency;
3. recommend suitable improvements, such as insulation, heat pumps, solar panels, double glazing and smart energy controls;
4. estimate potential energy savings and carbon reduction;
5. estimate the potential increase in property value.

The system should answer: *How green is my home today, what should I improve, how much energy and carbon could I save, and could this increase my property's value?*

Core rule. The score, savings, carbon and value figures come from documented models and calculations. The LLM only explains them and answers what-if questions; it must not calculate the score or invent estimates. For a what-if request, the relevant models and calculations must first be rerun for the proposed scenario, and the LLM then explains their outputs.

2 Green Home Score

Each property gets a score from 0 to 100, computed as a weighted sum of seven component sub-scores s_k, each from 0 to 100:

$$S = \sum_k w_k s_k$$

The weights w_k are fixed and are expressed as fractions in the formula (so $\sum_k w_k = 1$). How each sub-score is derived is the team's choice, but the mapping must be documented, and a better state must never give a lower sub-score. To reduce unintended double-counting, Energy efficiency should reflect the property's recorded energy rating (or a documented rating-based measure), while Energy consumption intensity should reflect annualised metered energy per square metre. If the same input affects more than one component, teams must justify this choice. Round the final score to the nearest integer for reporting and classification.

Component	Weight
Energy efficiency	25%
Energy consumption intensity (kWh per m^2)	20%
Insulation	15%
Heating system	15%
Windows and glazing	10%
Renewable energy	10%
Smart energy management	5%
Total	100%

Score	Classification
80–100	Highly efficient
65–79	Good
50–64	Moderate
30–49	Poor
0–29	Very poor

The system must show why a property received its score, including each component's contribution.

Example. Property P10001: rating E, high electricity and gas use, no solar, poor insulation, gas boiler, single glazing. Score 38/100, Poor.

3 Identify Poorly Performing Homes

Flag properties classified as Poor or Very poor and list their key issues (for example high energy use per m^2, poor insulation, old heating system, single glazing, no renewable generation).

Rank properties by **improvement potential**: the estimated percentage reduction in annual purchased energy achievable with the retrofits feasible for that property. For this metric, annual purchased energy is the sum of grid electricity imported and gas consumed, in kWh/year. Calculate improvement potential as $100 \times (E_{\text{baseline}} - E_{\text{after}})/E_{\text{baseline}}$, where both energy totals use the same accounting convention. If baseline energy is zero or unavailable, report the metric as unavailable and document the handling. Report electricity and gas changes separately, since a retrofit may reduce one while increasing the other. Solar panels reduce grid imports, not the home's underlying energy demand. A large house must not be classed as inefficient solely because its total consumption is higher: consider energy intensity and relevant property characteristics.

4 Retrofit Recommendations

Return a prioritised list of retrofits that are feasible for the property, not every retrofit available. Each item gives its suitability (High, Medium or Low), estimated saving and reason. The order should reflect how retrofits interact; for example, insulation before a heat pump. The impacts below are illustrative and refer to the quantity named in each row; they are not directly comparable or additive. Any retrofit catalogue used by the team takes precedence over these illustrative figures; teams must document its source and assumptions.

Retrofit	Illustrative impact
Roof or loft insulation	12–18% reduction in space-heating energy demand
Heat pump	15–25% reduction in purchased energy; electricity use may rise as gas use falls
Solar panels	20–35% reduction in grid electricity imports
Double glazing	8–12% reduction in space-heating energy demand
Smart thermostat	5–10% reduction in heating energy use

5 Energy Savings

Estimate the saving for each recommended retrofit and for the package as a whole, in kWh/year, separately for electricity and gas. Use a consistent baseline and estimate the post-retrofit consumption for each fuel. Show an increase in consumption as a negative saving. For a package, account for interactions between retrofits rather than simply adding their individual estimates.

Illustrative aggregate example. Baseline electricity use of 4,800 kWh/year plus gas use of 12,000 kWh/year gives 16,800 kWh/year of purchased energy. If a retrofit is estimated to reduce this combined amount by 15%, the aggregate saving is $16{,}800 \times 0.15 = 2{,}520$ kWh/year. This aggregate example does not determine the electricity and gas savings: the model must estimate and report the change for each fuel separately. The full package may, for example, have a net estimated saving of 5,500 kWh/year after accounting for interactions.

All savings are labelled as estimates, not guarantees.

6 Carbon Reduction

Convert energy savings to carbon using documented emissions factors for each fuel. Teams may use suitable factors from a reliable source, with units and source recorded:

$$\Delta C = \sum_{f \in \{\text{elec, gas}\}} \Delta E_f \times \text{EF}_f$$

where $\Delta E_f = E_{f,\text{baseline}} - E_{f,\text{after}}$ is the change in consumption of fuel f, in kWh/year, and EF_f is its factor in kg CO_2e per kWh. A positive ΔE_f is a saving; a negative value is an increase in consumption. A positive ΔC therefore represents an emissions reduction. Per-fuel calculation matters because a heat pump may lower gas use while raising electricity use.

7 Property Value Impact

Estimate the potential change in property value as a range, not a single number.

Example. Current value £350,000; rating E to C; indicative uplift 2–5%; potential value £357,000 to £367,500.

Describe it as an *estimated potential property-value impact based on historical and market relationships in the dataset prepared by the team*. Compare a consistent baseline with the proposed post-retrofit scenario (for example, predicted values for the same property before and after the proposed improvements, holding other property characteristics constant). Account for location, size, property type and market conditions. If the available observational data and method cannot establish a causal retrofit effect, describe the result as an indicative estimate rather than a guaranteed or proven causal increase.

8 Expected Outputs

Output	Example
Green Home Score	38/100
Efficiency classification	Poor
Recommended improvements	Insulation, heat pump, solar
Potential annual energy savings	5,500 kWh/year net, with electricity and gas changes reported separately
Potential annual carbon reduction	1.2 t CO_2e/year, calculated from the per-fuel changes
Potential property-value impact	+2–5% (indicative range)

Pipeline. Property, energy and sales data → Feature engineering → Green Home Score → Efficiency detector → Retrofit engine → Savings, carbon and value models → LLM explanation → Customer dashboard.

What-if flow. A what-if request updates the proposed retrofit scenario, reruns the relevant savings, carbon and value calculations, and passes those computed results to the LLM for explanation. The LLM must not invent scenario outputs.

Suggested methods. Score: explainable weighted scoring. Energy use: regression (XGBoost, LightGBM, random forest, neural network). Retrofits: classification or recommendation model. Carbon: deterministic calculation. Value: regression on property features, rating, retrofits and sales history. LLM: explains the score and recommendations, and answers what-if questions using values computed by the other components.

9 Dataset

Teams are expected to prepare the data needed for their solution. They may generate synthetic property, energy, retrofit and sales data, use suitable publicly available datasets that may be lawfully used, or combine both. Real personal or confidential customer data must not be used; public datasets must be used in accordance with their licences and applicable restrictions. The tables below suggest entities and fields. Teams may adapt them, provided their data supports the required outputs and functional requirements. Teams must document data sources, schema, assumptions and limitations, and the generation process and label-generation rules for synthetic data.

Teams must use property-level train, validation and test splits so that records for the same property do not appear in multiple splits. Where multiple properties belong to one customer, or customer memory is used, teams should split by customer to avoid leakage. Teams must also avoid using fields that would not be available at prediction time. Monthly energy usage should be annualised consistently using the available months; missing months or fields must be documented and handled explicitly rather than treated as zero by default.

Table	Suggested fields
Properties	property_id, customer_id (if applicable), property_type, age, floor_area, bedrooms, energy_rating, heating, insulation, glazing, solar, smart_controls, region, value
Energy usage	property_id, month, electricity_kwh, gas_kwh
Retrofit catalogue	retrofit, applicability conditions, indicative cost, typical impact
Emissions factors	fuel, kg_co2e_per_kwh, source
Retrofit history	property_id, retrofit, completion_date
Sales history	property_id, date, price, energy_rating_at_sale
Labels/reference estimates	improvement potential or ranking; retrofit feasibility/relevance; per-fuel savings; property-value impact, as applicable

Teams must provide or construct the reference labels or estimates needed to evaluate the required outputs, including improvement-potential rankings, retrofit feasibility/relevance, per-fuel savings, and property-value impact. These may be derived from observed data, documented rules, suitable published sources, or a simulator, as appropriate. Teams must document how the targets were obtained or generated. Synthetic labels must reflect plausible relationships with the input features rather than being assigned arbitrarily. If target values are generated by rules or a simulator, teams must disclose those rules and acknowledge that agreement with synthetic targets alone does not establish real-world predictive accuracy.

Teams must use documented emissions factors for all carbon estimates, including units and sources. Where appropriate factors are not included in their data, teams must source suitable factors and include them with the submission. The data schema must distinguish the current property-value estimate used as a baseline from historical sale prices, and identify fields unavailable at prediction time.

Because teams may use different datasets and label-generation methods, their numerical evaluation results may not be directly comparable. Judges will consider data quality, the validity and documentation of labels, held-out evaluation methodology, performance, and reproducibility.

Suggested public datasets (optional). Teams may find the following datasets useful:

- **Energy Performance of Buildings (EPC) Data:** Dataset link. Contains Energy Performance Certificate records for domestic properties in England and Wales, including energy ratings and property characteristics. The associated recommendations data may help teams develop retrofit recommendations and estimate potential improvements.
- **London Smart Meter Energy Consumption:** Dataset link. Contains electricity consumption readings at half-hourly intervals for a sample of 5,567 London households, collected between 2011 and 2014. It

can be used to develop energy-use profiles and analyse consumption patterns. The dataset is large, so teams may use a suitable subset.

- **National Energy Efficiency Data-Framework (NEED):** Dataset link. Provides household electricity and gas consumption statistics across different property and household characteristics. It can help teams establish consumption benchmarks and estimate typical energy use for different types of homes. The published tables are aggregated and do not provide individual-property records.
- **HM Land Registry Price Paid Data:** Dataset link. Contains residential property transaction prices, sale dates and property characteristics for England and Wales. It can support property-price analysis and the development of indicative property-value models.

These datasets cover only parts of the problem and may not be directly joinable at the property level. Teams may combine public datasets with synthetic data to meet the requirements. They must document their data sources, licensing conditions, assumptions and limitations, and must not present historical associations as proof of the causal effect of retrofits on property value.

10 Optional Extension: Customer Memory

Teams may give each customer a persistent memory, such as completed retrofits, stated constraints or preferences, and past recommendations with the customer's responses, and use it to adapt later recommendations. The design is open. Scores and estimates must still come from the models and calculations above. Teams that attempt this should show how recommendations change as memory builds up.

11 Evaluation

Teams must evaluate their solutions using documented train, validation and test splits where applicable. They must report their evaluation methodology, relevant metrics, baseline comparisons and limitations. The quality of the data, label-generation process, validation methodology and reproducibility will be considered alongside reported results.

Since teams may use different datasets and label-generation methods, numerical metrics from different submissions are not directly comparable and will not be used as a standalone ranking criterion.

Area	Evaluation criteria
Data quality	Plausibility of the data, documented sources and assumptions, label-generation methodology, treatment of missing data, and prevention of data leakage.
Green Home Score	Correct use of the fixed weights, documented and monotonic sub-score mappings, reproducibility, and explanation of component contributions.
ML and recommendations	Appropriateness of the approach, validation methodology, comparison with a simple baseline, quality and ranking of recommendations, and feasibility of proposed retrofits.
Energy savings and carbon reduction	Consistency and reproducibility of per-fuel calculations, treatment of interactions between retrofits, appropriate use of documented emissions factors, and comparison with held-out reference estimates where available.
Property-value impact	Transparency of the estimation method and assumptions, evaluation of the submitted range against held-out reference estimates where available, and appropriate treatment of confounding factors and causal limitations.
LLM explanations	Groundedness, factual accuracy, completeness, consistency with model outputs, and appropriate handling of missing or conflicting information and infeasible retrofits.
System performance	Responsiveness, reliability, reproducibility, and reasonable computational and LLM costs.

Scenario-based evaluation. During evaluation, organisers may test submissions using scenarios involving missing or conflicting property data, infeasible or interacting retrofits, increased electricity use after a heating-system change, and insufficient information to estimate savings or property-value impact.

LLM explanation evaluation. Organisers may evaluate explanations and what-if responses using held-out scenarios, including missing data, conflicting signals and infeasible retrofits. Exact evaluation scenarios need not be disclosed in advance. Targets are groundedness $\geq$90%, factual accuracy $\geq$90%, and unsupported claims <5%.

Baseline and reproducibility. Teams must compare their approach with an appropriate simple baseline on their own validation data and explain the results. The organisers may rerun submitted code and inspect the data, labels, calculations and outputs. Reported results that cannot be reproduced may affect the evaluation. Because datasets may differ across teams, judges will consider the data and label-generation methodology alongside reported metrics rather than treating raw metrics as directly comparable across all teams.

12 Scoring

Submissions will be evaluated across the following criteria. Teams should demonstrate their approach and provide evidence from their implementation and evaluation; the sophistication of a model alone does not determine its quality.

Criterion	Weight
Business relevance and usefulness of the recommendations	10%
Green Home Score and sustainability insight quality	15%
ML and recommendation quality (data quality, validation methodology and held-out metrics)	20%
Energy savings, carbon reduction and property-value estimation	15%
GenAI and LLM implementation (grounded explanations and what-if evaluation)	15%
Technical architecture (including data engineering, reproducibility and system performance)	10%
Explainability and Responsible AI	5%
UX and dashboard	5%
Innovation (including the optional extension)	5%
Total	100%

Evaluation will consider the quality and documentation of the team's data and labels, the validity of its held-out evaluation, and whether its outputs are reproducible. Since teams may use different datasets and label-generation methods, raw metrics will not be treated as directly comparable without considering these differences. Unsupported precision, arbitrary synthetic labels, or claims that exceed what the data can establish may reduce the score.

13 Submission

A. Working prototype (web app or dashboard)

B. Source code and data preparation/generation scripts, with run instructions; include dataset files where sharing is permitted

C. Trained models and inference code

D. Scoring method: documented sub-score mappings

E. LLM implementation: prompts, model or API details, evaluation approach

F. Architecture diagram

G. Evaluation report: the metrics in Section 11, on the validation set

H. Responsible AI note: privacy, bias, explainability, hallucination, human oversight, data limitations

I. Held-out test predictions and the associated data schema/label documentation